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$$\lim_{x\to\infty}(\sqrt{x^2+3x}+x)$$
$$= \lim_{x\to\infty}\frac{(\sqrt{x^2+3x}+x)(\sqrt{x^2+3x}-x)}{(\sqrt{x^2+3x}-x)}$$
$$= \lim_{x\to\infty}\frac{x^2+3x-x^2}{\sqrt{x^2+3x}-x}$$
$$\lim_{x\to+\infty}\frac{3x}{x-x} = \frac{3x}{0} = +\infty$$
$$\lim_{x\to-\infty}\frac{3x}{-x-x} = -\frac{3}{2}$$

References